

Session 3: Internet & Networking Basics.

Task 1: Protocol Full Forms and Uses

- **HTTP**
 - **Full Form:** Hypertext Transfer Protocol.
 - **Use:** The foundational protocol used for transmitting web pages and data over the internet from a server to a browser.
- **HTTPS**
 - **Full Form:** Hypertext Transfer Protocol Secure.
 - **Use:** The secure, encrypted version of HTTP that protects sensitive data (like passwords and credit cards) during transmission.
- **FTP**
 - **Full Form:** File Transfer Protocol.
 - **Use:** A standard network protocol used specifically for transferring files between a client and a server over the internet.
- **SMTP**
 - **Full Form:** Simple Mail Transfer Protocol.
 - **Use:** A protocol used by mail servers to send, relay, and deliver emails from one user to another.

Task 2: IPv4 vs. IPv6

The Difference:

IPv4 uses a 32-bit numerical address format (e.g., 192.168.1.1) which limits it to about 4.3 billion unique addresses. Conversely, **IPv6** uses a 128-bit hexadecimal address format (e.g., 2001:db8::ff00:42:8329) to provide an virtually infinite number of unique addresses for the growing number of global devices.

Task 3: nslookup google.com Result

When you run the command `nslookup google.com` in a computer terminal (Command Prompt or Terminal), the system queries the DNS server to find the IP address. A standard result looks like this:

```
Plaintext
```

```
Server: UnKnown
```

```
Address: 192.168.1.1
```

```
Non-authoritative answer:
```

Name: google.com

Addresses: 2404:6800:4009:831::200e

142.250.195.142

- **The IPv4 Address returned is:** 142.250.195.142
- **The IPv6 Address returned is:** 2404:6800:4009:831::200e
(Note: The exact IP addresses may vary slightly depending on your geographic location and the server you hit).

Task 4: HTTP Status Codes Meaning

- **200 OK:** The request was successful, and the server successfully delivered the requested data or webpage.
- **301 Moved Permanently:** The requested webpage or resource has been permanently assigned a new web address (URL).
- **403 Forbidden:** The server understands the request, but refuses to authorize it; you do not have permission to access this resource.
- **404 Not Found:** The server cannot find the requested webpage or file because it has been deleted or the URL was mistyped.
- **500 Internal Server Error:** A generic error message indicating that the server encountered an unexpected condition that prevented it from fulfilling the request.

Task 5: DNS Hierarchy (Using www.google.com)

The Domain Name System (DNS) is structured like an inverted tree. When resolving www.google.com, it is read from right to left by the system:

- **Root Level (.):** The invisible dot at the very end of a web address (implied as www.google.com.). It points to the Root Servers, which know where to find the Top-Level Domains.
- **TLD - Top-Level Domain (.com):** The highest level in the DNS hierarchy structure. It categorizes the type of domain (e.g., .com for commercial, .org for organizations).
- **Domain Level (google):** The unique name registered by an organization or individual. It identifies the specific entity owning the website.
- **Subdomain Level (www):** A subsection created under the main domain to organize specific services or sections of a website (e.g., www for the main web server, mail for email services).